

A Hybrid EfficientNet-Transformer with Attention Mechanism for Brain Tumor Classification from MRI Images

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ABSTRACT: Classification of brain tumors from MRI plays an important role in assisting the clinical process of diagnosis and treatment planning. In this work, a new deep learning model is proposed which uses EfficientNet-B0, a Transformer Encoder, and attention-based mechanism for classifying multiple brain tumors. EfficientNet-B0 is employed to obtain distinctive visual features from MRI images, while the Transformer encoder captures global context among different regions in the image. The use of an attention mechanism helps in improving tumor-specific features before classification. The model is tested on the dataset consisting of MRI brain tumors classified into four categories namely, glioma, meningioma, pituitary, and no-tumor. Based on experimental results, a validation accuracy of 99.02% is achieved by the model, proving its potential for automated brain tumor classification.

Keywords: Brain Tumor Classification; Magnetic Resonance Imaging; EfficientNet-B0; Transformer Encoder; Attention Mechanism; Deep Learning.

1. INTRODUCTION

Brain tumors constitute an important class of neurological disorders for which timely diagnosis is essential to guide treatment planning. MRI constitutes a widely-used technique for brain tumor evaluation owing to its ability to visualize soft tissues and various anatomical structures. However, MRI scan interpretation usually takes a significant amount of time and is prone to inter-observer differences, making automated and robust computer-aided diagnostic solutions necessary [1]. The latest developments in deep learning techniques significantly improved MRI-based brain tumor classification. Convolutional neural networks exhibited a high level of expertise in learning hierarchical visual features and found wide application in various tumor recognition and classification problems [2], [3]. The application of transfer learning based on pre-trained convolutional neural networks further boosted classification performance and decreased computing requirements [3]. Various convolutional neural network-based models were proposed for multi-class brain tumor classification; however, the most notable are the EfficientNet architectures, which achieved impressive classification performance by combining efficient scaling techniques [4], [5], [6], [7]. However, despite the advancements made possible by CNN architectures, these networks mainly focus on extracting local features and thus may fail to incorporate long-range context relations found in complex MRI imagery. Attention mechanisms were proposed to be introduced into deep learning frameworks in order to identify important visual features and improve discriminative power [8], [9]. In recent years, transformer-based architectures were demonstrated to learn global dependencies of visual features using self-attention mechanisms in order to learn better representations from medical images [10], [11].

The primary contributions of the proposed framework include:

1. Developing a hybrid EfficientNet Transformer architecture for MRI-based brain tumor classification.
2. Integration of an attention mechanism for feature refinement and improved discrimination.
3. Comparative evaluation against recent state-of-the-art deep learning methods.
4. Achievement of 99.02% validation accuracy on a multi-class brain tumor MRI dataset.

2. RELATED WORK

Deep learning techniques have become the dominant approach used in MRI brain tumor classifications due to their inherent ability to automatically discover discriminative image features. Traditional CNN architectures have proven effective in detecting and classifying brain tumors through the hierarchical extraction of spatial representations in MRI scans [1], [2]. Classification efficiency has been further increased using transfer learning techniques that make use of pre-trained models while requiring less training data [12]. The EfficientNet architectures has gained prominence due to its computational efficiency and optimal scaling strategy. Several studies indicate good performance of EfficientNet variations in multi-class brain tumor classification, implying better feature extraction capability and decreased model complexity [13], [14], [15], [16]. Moreover, attention-enabled EfficientNet architectures have shown better classification results through the focus on important brain tumor regions while learning image features [17].

Attention and Transformers' ability to model long-range dependencies has enabled improvements in medical imaging tasks. Inclusion of attention mechanisms in medical image analysis models through additional modules has led to improved feature representations and higher classification performance [18], [19]. Similarly, Transformer models rely on self-attention operations to incorporate global context information and produce state-of-the-art results in MRI brain tumor classification [10], [13]. Recently, hybrid architectures combining CNNs, attention mechanisms, and Transformers have shown promise in medical image classification, as a result of synergy between local feature extraction and global feature representation approaches [20], [21], [22].

3. Proposed Methodology

The proposed framework integrates EfficientNet-B0, a transformer encoder, and the attention method to perform multi-class brain tumor classification on MRIs. The whole process involves image preprocessing, feature extraction using EfficientNet, context-based learning using the transformer, refining extracted features using attention, and finally performing classification. These are illustrated Fig. 1.

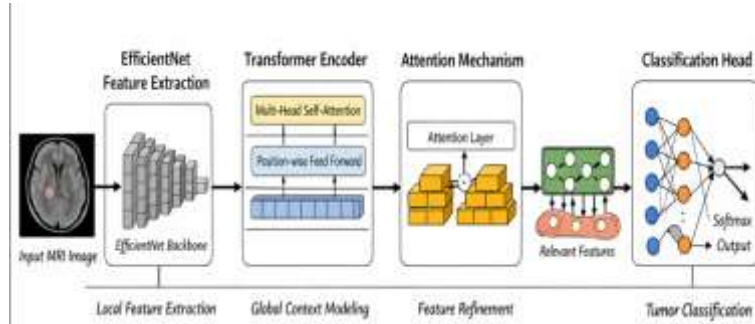


Fig. 1. Proposed hybrid EfficientNet-Transformer framework for brain tumor classification.

3.1 Image Preprocessing

The tests were carried out on the Brain Tumor MRI Dataset [24]. The dataset includes four MRI images classes: glioma, meningioma, pituitary, and no-tumor. The experiments used an 80:20 train/validation split to evaluate the framework's accuracy.

The distribution of the number of MRI images used in this experiment. These images are distributed into four categories in equal proportions, namely glioma, meningioma, pituitary, and no-tumor. Out of these 7,200 images, 5,600 are used for training, while the rest 1,600 are set aside for testing.

MRI images have various inconsistencies in terms of intensity, contrast, and other factors. To counteract these effects, all images were resized to 224×224 in size, which is the requirement of EfficientNet-B0. Normalizing the images was also done to increase the stability of the model training process:

$$I_{norm} = \frac{I - \mu}{\sigma}$$

where I represents the input MRI image, while μ and σ denote the mean and standard deviation, respectively.

3.2 EfficientNet-Based Feature Extraction

In this proposed framework, EfficientNet-B0 has been used for extracting the feature representation from MRI images. In fact, efficient representations are generated by this network, which contain the relevant structure and texture information of the image. These feature maps are then fed into the transformer encoder for global context learning.

The extracted feature representation is given by:

$$F = \text{EfficientNet}(I_{norm})$$

where F denotes the feature map generated by the EfficientNet backbone.

3.3 Transformer-Based Contextual Learning

Although convolutional layers effectively capture local spatial information, they may not adequately model long-range dependencies within MRI images. To overcome this limitation, the extracted feature maps are transformed into feature tokens and processed by a Transformer encoder.

The self-attention mechanism computes relationships among feature representations according to:

$$A_{cross} = \text{softmax}\left(\frac{F_{cnn}F_{swin}^T}{\sqrt{d}}\right)$$

$$F_{fusion} = A_{cross}F_{swin} + F_{cnn}$$

The obtained feature maps from EfficientNet are transformed into tokens and embedded into the feature embeddings space to be used as input to the Transformer encoder. By using multi-head self-attention, the network can learn inter-dependencies of distant regions of the MRI scan.

3.4 Attention-Guided Feature Refinement

Embeddings obtained from the encoder of the Transformer model are then sent to an attention-based processing step. Attention processing step emphasizes important features related to tumors, and improves the learned representation before classification. The improved representation of the feature can be formulated as:

$$F_{att} = \sigma(WF + b) \odot F$$

Subsequently, the learned contextual features are sent through an attention layer which adaptively assigns weights to the obtained features based on their importance for tumor classification. The attention mechanism enables the network to emphasize discriminative tumor features.

3.5 Classification and Model Training

The refined features are mapped to the output classes through fully connected layers. A Softmax function is then applied to generate the probability score for each tumor category. The resulting probability distribution is given by:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

Model learning is guided by the categorical cross-entropy objective:

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

3.6 Training Hyperparameters

Performance of the EfficientNet-Transformer model is dependent on certain hyperparameters that have been used during the training process. Proper parameter selection is essential in ensuring stable training, quick convergence, and improved accuracy of classification. The training hyperparameters used in the training process are detailed in Table 1.

Table 1. Training Hyperparameters

Parameter	Value
Optimizer	Adam

Learning Rate	0.0001
Batch Size	32
Epochs	25
Input Size	224×224

4. RESULTS AND ANALYSIS

4.1 Training Convergence Analysis

The learning process of the suggested EfficientNet-Transformer architecture was conducted in 25 training epochs. According to Fig.2, there were fast growths in the training accuracy and the validation accuracy in the first few epochs, with the convergence being achieved after six epochs. In particular, the last training accuracy reached 99.82%, whereas the validation accuracy became equal to 99.02%.

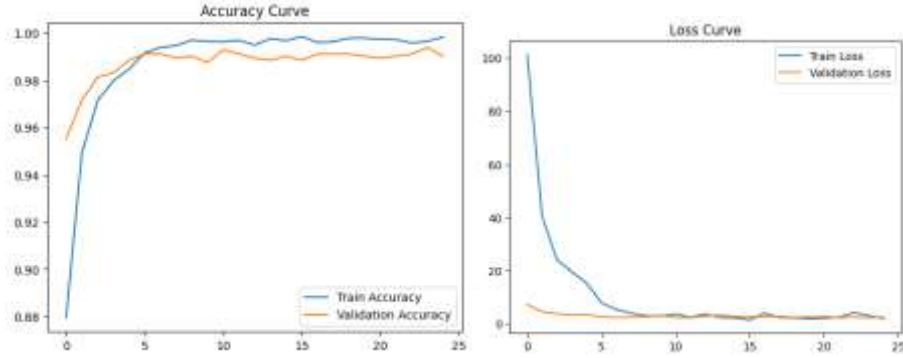


Fig. 2. Training and validation performance of the proposed model.

4.2 Confusion Matrix Analysis

The confusion matrix on the validation data set is shown in Figure 3. The proposed framework showed its robust performance on most of the MRI samples with correct identification of each sample in all categories. A slight misclassification was noted for the Glioma and Meningioma classes, because of their similar imaging features, while the No Tumor class had no misclassification. The results demonstrate the effectiveness of the proposed model in the classification of MRI images of brain tumors with multiple classes.

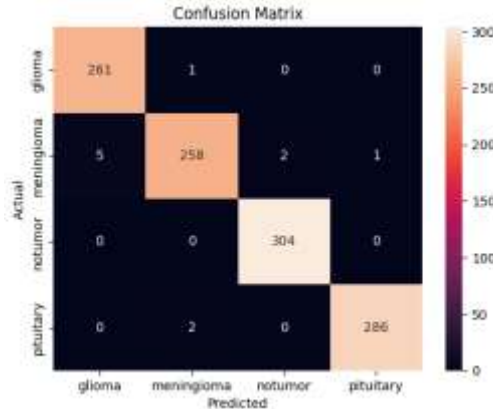


Fig. 3. Confusion matrix of brain tumor classification results.

4.3 Comparative Performance Analysis

The classification results of the proposed framework were compared to the classification results reported by the recent deep learning approaches for brain tumour MRI classification to assess its performance. The comparative results are summarised in Table 2.

Table-2 Performance Comparison with State-of-the-Art Brain Tumor Classification Methods

Reference	Method	Dataset	Accuracy (%)
Shah et al. [7]	Fine-Tuned EfficientNet	Brain MRI	98.57
Preetha et al. [18]	EfficientNet-B4	Brain MRI	98.74
Hassan and Ghadiri [4]	EfficientNetV2 Transfer Learning	Brain MRI	98.81

Celik et al. [8]	Ensemble Attention Network	Brain MRI	98.90
Panigrahi et al. [11]	Transfer Learning + Self-Attention	Brain MRI	98.94
Vimala et al. [21]	Hybrid Deep Learning Model	Brain MRI	98.96
Proposed Method	EfficientNet-B0 + Transformer + Attention	Brain MRI Dataset	99.02

4.4 Ablation Analysis

The study used ablation to evaluate the impact of each element of the proposed architecture. Different architectures of the model were analysed to study the impact of the attention module and the Transformer encoder on classification accuracy. The results are presented in Table 3 below.

Table 3. Ablation Analysis of Different Model Configurations.

Configuration	Accuracy (%)
EfficientNet-B0	97.8
EfficientNet-B0 + Attention	98.4
EfficientNet-B0 + Transformer	98.6
Proposed Hybrid Model	99.02

The ablation results prove that the Transformer encoder and the attention mechanism both help to achieve better classification accuracy, and the hybrid full framework has the highest accuracy.

4.5 Discussion

From the experiment results above, it is shown that the proposed approach has made use of the complementary roles of the EfficientNet-B0 architecture, the Transformer encoder, and the attention model. EfficientNet-B0 can extract representative visual features from MRI images, whereas the Transformer encoder learns the interactions between feature maps that may not be captured effectively using only convolution operations. With the help of the attention model, the important features that indicate the presence of tumor areas have been focused upon, hence making feature discrimination more efficient.

5. CONCLUSION AND FUTURE WORK

This study proposes a hybrid approach based on the combination of EfficientNet-B0, Transformer encoder, and the attention mechanism to classify multiple types of brain tumors using MRIs. The suggested approach combines the capabilities of convolutional feature extraction, modelling of global dependencies between different regions, and enhancing the discriminative features using attention mechanisms. The EfficientNet-B0 model is used to create efficient feature representations using the MRI images, and the transformer encoders are used to capture the inter-regional interactions between regions in order to provide context information. The attention module then makes use of these representations and highlights relevant patterns that help in characterizing the tumors. Validation testing of the framework using the Brain Tumor MRI Dataset produced a 99.02% accuracy result, which proves the efficiency of the proposed approach for multiclass classification purposes. It performed effectively for gliomas, meningiomas, pituitary tumors, and no-tumors. Future work can involve validation of the framework using different and larger datasets along with explainable lightweight models.

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